

Project 3

Botnet PCAP Trace Analysis Using Wireshark

Objective

- To use Wireshark to analyze network packets of a botnet.

Description

- Students may work individually or form a team of up to four members to complete the project. While teamwork is allowed, students are encouraged to complete the project on their own.
- All students must submit their own answers to this project.
- Graduate students are required to answer all questions.
- Undergraduate students may select questions totaling 24 points to answer. The maximum score for an undergraduate student will be capped at 24, regardless of how many questions are attempted.

Submission

- All students must complete Project 3, which is listed under the Quiz category;
- All students must also submit an analysis report (using this [template](#)) explaining how the results were derived from the given trace. Students working in the same team may submit an identical report; however, each student must upload the report individually.
- For your answer to each question in the analysis report, please include the screenshot of the relevant code snippet and your analysis.

Pcap File and Additional Information

- You can download the pcap file of the network traffic from this [link](#).
- You will find the source code of the bot in HW3 helpful.
- You will find the IRC [protocol](#) useful for this project.

The Tool

- You are recommended to use Wireshark to complete this project.
- You may want to use the Wireshark's "filter" functions to expediate your analysis.

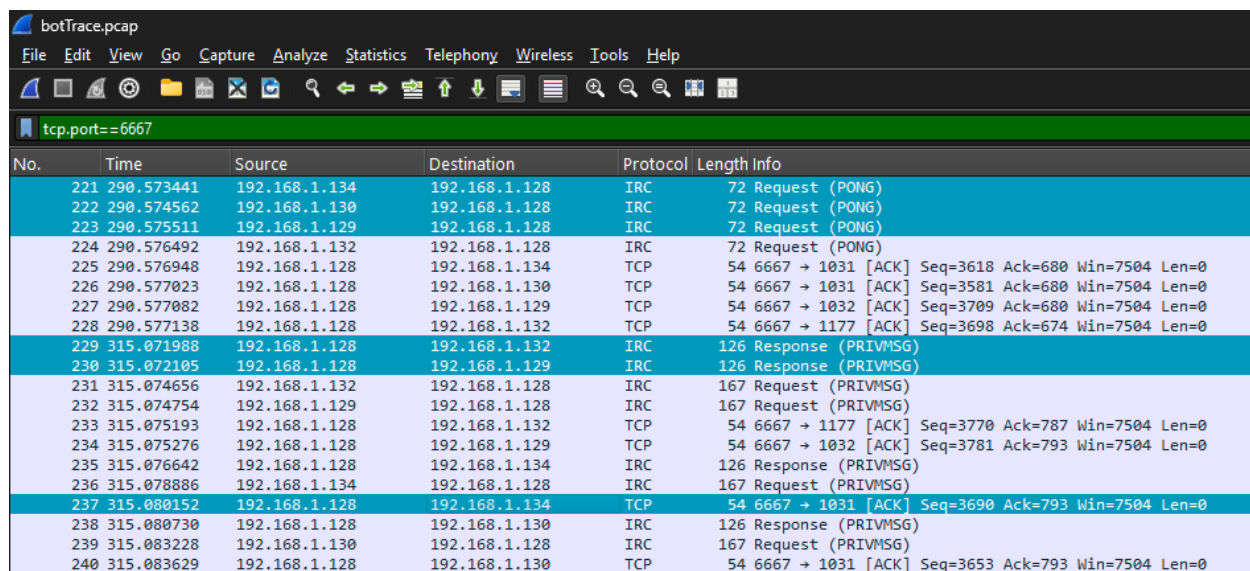
Your Name: Bhuwan Pandey

UID: U01155118

Q1. Which of the following IP addresses is the botnet's command-and-control (C2) server?

192.168.1.128
192.168.1.129
192.168.1.132
192.168.1.134

Answer: 192.168.1.128 seems like the botnet's C2 server as multiple hosts (192.168.1.129, .130, .132, .134) are communicating with 192.168.1.128
Also, we can see many IRC PRIVMSG, PING, and PONG messages centered around 192.168.1.128.



No.	Time	Source	Destination	Protocol	Length	Info
221	290.573441	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
222	290.574562	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
223	290.575511	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
224	290.576492	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
225	290.576948	192.168.1.128	192.168.1.134	TCP	54	6667 → 1031 [ACK] Seq=3618 Ack=680 Win=7504 Len=0
226	290.577023	192.168.1.128	192.168.1.130	TCP	54	6667 → 1031 [ACK] Seq=3581 Ack=680 Win=7504 Len=0
227	290.577082	192.168.1.128	192.168.1.129	TCP	54	6667 → 1032 [ACK] Seq=3709 Ack=680 Win=7504 Len=0
228	290.577138	192.168.1.128	192.168.1.132	TCP	54	6667 → 1177 [ACK] Seq=3698 Ack=674 Win=7504 Len=0
229	315.071988	192.168.1.128	192.168.1.132	IRC	126	Response (PRIVMSG)
230	315.072105	192.168.1.128	192.168.1.129	IRC	126	Response (PRIVMSG)
231	315.074656	192.168.1.132	192.168.1.128	IRC	167	Request (PRIVMSG)
232	315.074754	192.168.1.129	192.168.1.128	IRC	167	Request (PRIVMSG)
233	315.075193	192.168.1.128	192.168.1.132	TCP	54	6667 → 1177 [ACK] Seq=3770 Ack=787 Win=7504 Len=0
234	315.075276	192.168.1.128	192.168.1.129	TCP	54	6667 → 1032 [ACK] Seq=3781 Ack=793 Win=7504 Len=0
235	315.076642	192.168.1.128	192.168.1.134	IRC	126	Response (PRIVMSG)
236	315.078886	192.168.1.134	192.168.1.128	IRC	167	Request (PRIVMSG)
237	315.080152	192.168.1.128	192.168.1.134	TCP	54	6667 → 1031 [ACK] Seq=3690 Ack=793 Win=7504 Len=0
238	315.080730	192.168.1.128	192.168.1.130	IRC	126	Response (PRIVMSG)
239	315.083228	192.168.1.130	192.168.1.128	IRC	167	Request (PRIVMSG)
240	315.083629	192.168.1.128	192.168.1.130	TCP	54	6667 → 1031 [ACK] Seq=3653 Ack=793 Win=7504 Len=0

Q2. Identify the port number used by the botnet's command-and-control (C2) server to control bots.

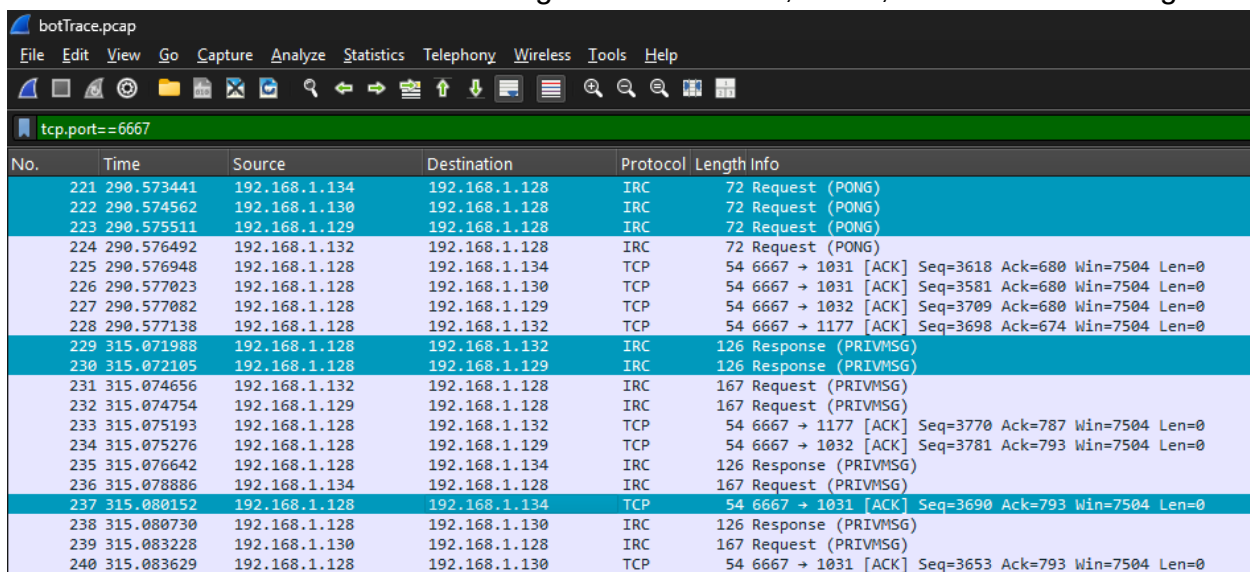
6667, 80, 443, 53

Answer: In reference to above screenshot (last highlighted row) and the reason being botnet using standard IRC on port 6667, we can clearly acknowledge that the port number used was 6667.

Q3. What application-layer protocol is used by the botnet's command-and-control (C2) channel?

IRC, HTTP, HTTPS, FTP

Answer: IRC is used as application-layer protocol because we can clearly see IRC shows in Protocol column and message like PRIVMSG, PING, PONG are exchanged.



No.	Time	Source	Destination	Protocol	Length	Info
221	290.573441	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
222	290.574562	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
223	290.575511	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
224	290.576492	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
225	290.576948	192.168.1.128	192.168.1.134	TCP	54	6667 → 1031 [ACK] Seq=3618 Ack=680 Win=7504 Len=0
226	290.577023	192.168.1.128	192.168.1.130	TCP	54	6667 → 1031 [ACK] Seq=3581 Ack=680 Win=7504 Len=0
227	290.577082	192.168.1.128	192.168.1.129	TCP	54	6667 → 1032 [ACK] Seq=3709 Ack=680 Win=7504 Len=0
228	290.577138	192.168.1.128	192.168.1.132	TCP	54	6667 → 1177 [ACK] Seq=3698 Ack=674 Win=7504 Len=0
229	315.071988	192.168.1.128	192.168.1.132	IRC	126	Response (PRIVMSG)
230	315.072105	192.168.1.128	192.168.1.129	IRC	126	Response (PRIVMSG)
231	315.074656	192.168.1.132	192.168.1.128	IRC	167	Request (PRIVMSG)
232	315.074754	192.168.1.129	192.168.1.128	IRC	167	Request (PRIVMSG)
233	315.075193	192.168.1.128	192.168.1.132	TCP	54	6667 → 1177 [ACK] Seq=3770 Ack=787 Win=7504 Len=0
234	315.075276	192.168.1.128	192.168.1.129	TCP	54	6667 → 1032 [ACK] Seq=3781 Ack=793 Win=7504 Len=0
235	315.076642	192.168.1.128	192.168.1.134	IRC	126	Response (PRIVMSG)
236	315.078886	192.168.1.134	192.168.1.128	IRC	167	Request (PRIVMSG)
237	315.080152	192.168.1.128	192.168.1.134	TCP	54	6667 → 1031 [ACK] Seq=3690 Ack=793 Win=7504 Len=0
238	315.080730	192.168.1.128	192.168.1.130	IRC	126	Response (PRIVMSG)
239	315.083228	192.168.1.130	192.168.1.128	IRC	167	Request (PRIVMSG)
240	315.083629	192.168.1.128	192.168.1.130	TCP	54	6667 → 1031 [ACK] Seq=3653 Ack=793 Win=7504 Len=0

Q4. Identify bots' IP addresses.

192.168.1.128
192.168.1.129
192.168.1.130
192.168.1.132
192.168.1.134

Answer: 192.168.1.129, 192.168.1.130, 192.168.1.132, 192.168.1.134 are the bot's IP addresses as they are observed communicating with the C2 server (192.168.1.128), indicating they are compromised hosts (bots).

No.	Time	Source	Destination	Protoc	Length	Info
39	62.346440	192.168.1.132	192.168.1.128	TCP	54	1177 → 6667 [ACK] Seq=51 Ack=59
33	60.253969	192.168.1.132	192.168.1.128	TCP	54	1177 → 6667 [ACK] Seq=1 Ack=1 W
31	59.973189	192.168.1.132	192.168.1.128	TCP	62	1177 → 6667 [SYN] Seq=0 Win=642
11750...	157197.271637	192.168.1.130	192.168.1.128	IRC	391	Request (PRIVMSG)
11750...	157197.262804	192.168.1.132	192.168.1.128	IRC	385	Request (PRIVMSG)
11750...	157197.232706	192.168.1.134	192.168.1.128	IRC	390	Request (PRIVMSG)
11750...	157197.210817	192.168.1.129	192.168.1.128	IRC	390	Request (PRIVMSG)
11750...	157191.308295	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
11750...	157191.305645	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
11750...	157191.304514	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
11750...	157191.303094	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
11750...	157096.293225	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
11750...	157096.291879	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
11750...	157096.290184	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
11750...	157096.288585	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
11750...	156999.525435	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
11750...	156999.524236	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
11750...	156999.523220	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
11750...	156999.522216	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
11749...	156904.286936	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
11749...	156904.285709	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
11749...	156904.284750	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)
11749...	156904.283474	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
11749...	156809.107659	192.168.1.130	192.168.1.128	IRC	72	Request (PONG)
11749...	156809.103145	192.168.1.134	192.168.1.128	IRC	72	Request (PONG)
11749...	156809.102241	192.168.1.132	192.168.1.128	IRC	72	Request (PONG)
11749...	156809.101995	192.168.1.129	192.168.1.128	IRC	72	Request (PONG)

Q5. What is the most likely password used by the botmaster to log in to the bots?

12345, password, ATtaCK!, test

Answer: The password is *test* as botmaster issues *.login 12345*, and all bots respond with “Password accepted,” confirming the password. Below image shows partial result after following one of the IRC packet.

```

:jjzhang!jjzhang@2401382F.247B6343.CD29BD2C.IP PRIVMSG #test :.login 12345
PRIVMSG #test :[MAIN]: Password accepted.
:[RX]|732143!nezwfb!w@192.168.1.130 PRIVMSG #test :[MAIN]: Password accepted.
:[RX]|320163!ytcpcfoqv@192.168.1.134 PRIVMSG #test :[MAIN]: Password accepted.
:[RX]|847125!ztzxroozq@192.168.1.129 PRIVMSG #test :[MAIN]: Password accepted.
:jjzhang!jjzhang@2401382F.247B6343.CD29BD2C.IP PRIVMSG #test :.sysinfo

```

Q6. What is the most likely version of the operating system on the compromised bots?

Windows XP (5.1, Build 2600)

Windows 2000 (5.1, Build 2600)

Windows NT (5.1, Build 2600)

None of above

Answer: The most likely version of the operating system on the compromised bots is *Windows XP (5.1, Build 2600)* as all bots report the same OS version.

```
[RX][847125]zxr00sq@192.168.1.129 PRIVMSG #test :[SYSINFO]: [CPU]: 2775MHz. [RAM]: 261,616KB total, 261,616KB free. [Disk]: 7,325,608KB total, 2,598,892KB free. [OS]: Windows XP (5.1, Build 2600). [Sysdir]: C:\WINDOWS\Syme]: ga-a60f1jffif65.localdomain (192.168.1.129). [Current User]: junjie. [Date]: 29:Oct:2007. [Time]: 12:01:05. [Uptime]: 0d 0h 28m.
[RX][320163]ytcqfoq@192.168.1.134 PRIVMSG #test :[SYSINFO]: [CPU]: 2766MHz. [RAM]: 261,616KB total, 261,616KB free. [Disk]: 7,325,608KB total, 2,577,824KB free. [OS]: Windows XP (5.1, Build 2600). [Sysdir]: C:\WINDOWS\Syme]: ga-a60f1jffif65.localdomain (192.168.1.134). [Current User]: junjie. [Date]: 29:Oct:2007. [Time]: 12:00:55. [Uptime]: 0d 0h 27m.
[RX][732143]nezwfbi@192.168.1.130 PRIVMSG #test :[SYSINFO]: [CPU]: 2775MHz. [RAM]: 261,616KB total, 261,616KB free. [Disk]: 7,325,608KB total, 2,682,224KB free. [OS]: Windows XP (5.1, Build 2600). [Sysdir]: C:\WINDOWS\Syme]: ga-a60f1jffif65.localdomain (192.168.1.130). [Current User]: junjie. [Date]: 29:Oct:2007. [Time]: 12:05:56. [Uptime]: 0d 0h 14m.
PING :irc.me.com
```

Q7. Which of the following commands is used by the botmaster to scan the network of compromised hosts (bots)?

.advscan, .sysinfo, .login, .flusharp

Answer: Following up on a stream, we can see that .advscan is used to scan the network:

```
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.advscan netbios 100 5 120 -b -r
```

Q8. All compromised hosts (bots) use the same Microsoft Windows Product ID CD Key.

True/False

Answer: True, because as we can see on the stream follow up, all the bots return the same Product ID for the `.getcdkeys` command.

```
:jjzhang!jjzhang@24013B2F.247B6343.CD298D2C.IP PRIVMSG #test :.getcdkeys
PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
PRIVMSG #test :[CDKEYS]: Search completed.
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[CDKEYS]: Search completed.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[CDKEYS]: Search completed.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[CDKEYS]: Search completed.
```

Q9. The botmaster issued the command `dns www.sina.com.cn` instructing each bot to perform a DNS query for that domain. Which bots followed this instruction and **actually** sent the DNS query?

192.168.1.129

192.168.1.130

192.168.1.132

192.168.1.134

Answer: The bots following the instruction and actually sent the DNS query (even though the resolution failed) are 192.168.1.129, 192.168.1.130, and 192.168.1.134 as we can see in the below image of the same follow up stream result.

```
:jjzhang!jjzhang@24013B2F.247B6343.CD298D2C.IP PRIVMSG #test :.dns www.sina.com.cn
PRIVMSG #test :[DNS]: Couldn't resolve hostname.
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[DNS]: Couldn't resolve hostname.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[DNS]: Couldn't resolve hostname.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[DNS]: Couldn't resolve hostname.
PING :irc.me.com
```

Q10. Which of the following commands were issued by the attacker to the bots?

.flusharp
.getcdkeys
.procs
.getclip
.netinfo

Answer: All of the mentioned commands were issued by the attacker to the bots because they all seem to appear in the stream.

```
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.flusharp
PRIVMSG #test :[FLUSHDNS]: ARP cache flushed.

:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[FLUSHDNS]: ARP cache flushed.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[FLUSHDNS]: ARP cache flushed.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[FLUSHDNS]: ARP cache flushed.
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.getcdkeys
PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
PRIVMSG #test :[CDKEYS]: Search completed.

:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[CDKEYS]: Search completed.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[CDKEYS]: Search completed.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :Microsoft Windows Product ID CD Key: (55274-023-2812393-22001).
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[CDKEYS]: Search completed.
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.getclip
PRIVMSG #test :-[Clipboard Data]-
PRIVMSG #test :Scope:Link

:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :-[Clipboard Data]-
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :Scope:Link
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :-[Clipboard Data]-
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :Scope:Link
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :-[Clipboard Data]-
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :Scope:Link
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.procs
PRIVMSG #test :[PROC]: Listing processes:
PRIVMSG #test :System (4)
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[PROC]: Listing processes:
```

```

:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.netinfo
PRIVMSG #test :[NETINFO]: [Type]: LAN (LAN Connection). [IP Address]: 192.168.1.132. [Hostname]: 192.168.1.132.
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[NETINFO]: [Type]: LAN (LAN Connection). [IP Address]: 192.168.1.1
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[NETINFO]: [Type]: LAN (LAN Connection). [IP Address]: 192.168.1.1
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[NETINFO]: [Type]: LAN (LAN Connection). [IP Address]: 192.168.1.1
:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.id
PRIVMSG #test :[MAIN]: Bot ID: rx01.
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[MAIN]: Bot ID: rx01.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[MAIN]: Bot ID: rx01.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[MAIN]: Bot ID: rx01.
PING :irc.me.com

```

Q11. The botmaster issued the command ***.advscan netbios 100 5 120 -b -r***, instructing each bot to perform TCP SYN scanning of port 139 (NetBIOS) against hosts in the same subnet. Bots should send a large number of TCP SYN packets to that subnet, causing a large number of failed TCP connections. Sometimes, bots do not follow the command. Please identify bots that **did not** execute this command.

192.168.1.129
 192.168.1.130
 192.168.1.132
 192.168.1.134

Answer: 192.168.1.32 did not execute this command because it didn't simply reply, as we can in the below stream.

```

:jjzhang!jjzhang@24013B2F.247B6343.CD29BD2C.IP PRIVMSG #test :.advscan netbios 100 5 120 -b -r
PRIVMSG #test :[SCAN]: Random Port Scan started on 192.168.x.x:139 with a delay of 5 seconds for 120 minutes using 100 threads.
:[RX]|847125!ztxroozq@192.168.1.129 PRIVMSG #test :[SCAN]: Random Port Scan started on 192.168.x.x:139 with a delay of 5 seconds for 120 minutes using 100 threads.
:[RX]|320163!ytcqfoqv@192.168.1.134 PRIVMSG #test :[SCAN]: Random Port Scan started on 192.168.x.x:139 with a delay of 5 seconds for 120 minutes using 100 threads.
:[RX]|732143!nezwfblw@192.168.1.130 PRIVMSG #test :[SCAN]: Random Port Scan started on 192.168.x.x:139 with a delay of 5 seconds for 120 minutes using 100 threads.
PING :irc.me.com

```

Q12. Which of the following bots likely issued ARP scanning to discover active IP addresses in the local area network?

192.168.1.129
 192.168.1.130
 192.168.1.132
 192.168.1.134

Answer: IP address 192.168.1.130 and 192.168.1.134 likely issued ARP scanning because the two has sent many ARP requests.

botTrace.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protoc	Length	Info
2730	1352.503783	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2731	1352.503829	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2735	1352.552736	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2736	1352.552826	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2737	1352.552880	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2738	1352.552932	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2739	1352.613055	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2740	1352.613144	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2741	1352.613192	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2742	1352.613236	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2747	1352.661516	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2748	1352.661595	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2749	1352.661642	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2750	1352.722439	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2751	1352.722521	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2752	1352.722568	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2763	1352.882596	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2764	1352.882679	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2765	1352.882727	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2766	1352.882783	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2767	1352.941155	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2768	1352.941237	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2769	1352.941451	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2770	1352.941492	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2775	1352.991912	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2776	1352.992005	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2777	1352.992052	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2778	1353.050362	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2779	1353.050503	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2780	1353.050567	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2784	1353.101352	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2785	1353.101461	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2786	1353.101528	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2787	1353.101588	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2788	1353.101651	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2789	1353.159846	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2790	1353.159936	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2791	1353.160004	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2792	1353.160065	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2793	1353.160126	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2794	1353.160191	VMware_d0:6c:d0	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.130
2799	1353.210624	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2800	1353.210704	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134
2801	1353.210750	VMware_79:f3:f5	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.134

As we can see in the Info column, the info with IP value suffixed .130 and .134 appears abundantly.